

**PES UNIVERSITY, BANGALORE**  
Department of Computer Science and Engineering

**Jackfruit Phase-1**

**Software Test Plan (STP)**  
**for the Railway Reservation System**

Version 1.0  
Date: 16-Sep-2026

**SUBMITTED BY**

| S.No | Name                  | SRN           |
|------|-----------------------|---------------|
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## Revision History

| Name    | Date        | Reason For Changes                  | Version |
|---------|-------------|-------------------------------------|---------|
| Initial | 16-Sep-2026 | Initial draft of Software Test Plan | 1.0     |
|         |             |                                     |         |

# 1. Introduction

## 1.1 Purpose

This Software Test Plan (STP) defines the objectives, scope, approach, resources, schedule and responsibilities for testing the Railway Reservation System, a console-based ticket reservation application developed in C++17 as the Mini-Project for the Software Engineering course, 5th Semester, PES University. It ensures that every functional and non-functional requirement defined in the Software Requirements Specification (SRS) is verified before submission.

## 1.2 Scope

Testing covers all passenger-facing features (registration, login, train search, seat availability, booking, payment, cancellation/refund, PNR enquiry, booking history, e-ticket generation) and all administrator features (train schedule management, fare and quota management, report generation), together with the security and performance requirements defined in SRS Sections 6.1 and 6.3.

Out of scope: real payment-gateway integration, multi-machine concurrent access, and ATM/network hardware, since the system is a single-user, offline console application (see SRS Section 1.3).

## 1.3 References

- Railway Reservation System SRS, Jackfruit Phase-1, PES University, 06-Sep-2026.
- SE Mini-Project Deliverables Part-1, PES University, ETA 28-Sep-2026.
- ISO/IEC/IEEE 29148:2018 - Systems and Software Engineering: Requirements Engineering.
- Software Architecture & Design Specification, Jackfruit Phase-1 (companion document).

## 1.4 Definitions

| Term           | Definition                                                                         |
|----------------|------------------------------------------------------------------------------------|
| PNR            | Passenger Name Record - unique 10-digit booking identifier.                        |
| SRS            | Software Requirements Specification.                                               |
| RAC            | Reservation Against Cancellation.                                                  |
| RTM            | Requirement Traceability Matrix.                                                   |
| FR / NFR / SEC | Functional / Non-Functional / Security requirement identifiers as used in the SRS. |
| UT             | Unit Test case identifier.                                                         |

# 2. Test Items

The following internal modules (SRS Section 2.1) are the items under test:

- AuthManager - registration, login, session and logout logic
- TrainSearch - train/fare lookup by source, destination and date
- SeatInventory - live seat occupancy and soft-lock handling
- BookingManager - booking creation, PNR assignment, Confirmed/Waitlisted/RAC status
- PaymentModule - fare calculation and simulated payment processing
- CancellationManager - cancellation and refund calculation
- PNRService / HistoryService - PNR status lookup and booking history
- TicketPrinter - e-ticket generation and export
- AdminTrainManager / AdminFareManager - schedule, fare and quota management
- ReportGenerator - occupancy and revenue reports
- FileStore - persistence of trains.dat, users.dat, bookings.dat

### 3. Features to be Tested

The features to be tested are mapped to their corresponding SRS requirement IDs. Detailed test-level traceability is provided in Section 13.

| Feature                            | SRS Requirement IDs            |
|------------------------------------|--------------------------------|
| User Registration & Authentication | FR-01 to FR-04, SEC-01, SEC-04 |
| Train Search                       | FR-05 to FR-07, NFR-01         |
| Seat Availability & Booking        | FR-08 to FR-13, NFR-02         |
| Payment Processing                 | FR-14 to FR-16                 |
| Cancellation & Refund              | FR-17 to FR-19                 |
| PNR Status & Booking History       | FR-20 to FR-22                 |
| E-Ticket Generation                | FR-23, FR-24                   |
| Admin: Train & Fare Management     | FR-25 to FR-28                 |
| Admin: Reports                     | FR-29, FR-30                   |
| Performance                        | NFR-01 to NFR-05               |
| Security                           | SEC-01 to SEC-05               |

### 4. Features Not to be Tested

- Real payment-gateway integration - the SRS specifies a simulated Payment Module only (SRS 1.3).
- Multi-user, multi-machine concurrent access - explicitly out of scope (SRS 2.5).
- Underlying OS console I/O and C++ Standard Library internals - assumed correct.
- Third-party CI tooling internals (Jenkins, SonarQube) beyond pipeline integration checks.

## 5. Test Approach / Strategy

### 5.0 Levels and Types

#### Levels:

- Unit tests - individual classes (AuthManager, BookingManager, SeatInventory, PaymentModule, CancellationManager) using GoogleTest, run via the Jenkins CI pipeline.
- Integration tests - interactions between BookingManager, SeatInventory, PaymentModule and FileStore.
- System tests - end-to-end console workflows from Main Menu through booking, payment, cancellation and admin flows.
- Acceptance tests - scenario walkthroughs against the SRS use cases (UC-01 to UC-13) with course evaluators.

#### Types:

- Functional testing - verifies each FR behaves as specified.
- Non-functional / performance testing - verifies NFR-01 to NFR-05 (response time, memory footprint).
- Security testing - verifies SEC-01 to SEC-05 (see Section 5.1 below).
- Regression testing - re-run of the GoogleTest suite on every pull request via Jenkins.
- Usability testing - informal walkthrough of console prompts and error messages by non-author team members.

Entry Criteria: A buildable C++17 codebase (via CMake), GoogleTest suite compiling, and test data files available.

#### Exit Criteria:

- 100% of planned test cases executed.

- No critical defects remain open.
- All high-priority requirements are covered by at least one passing test case.
- Required security (Section 5.1) and non-functional/performance tests are completed.
- Requirement traceability (Section 13) is reviewed and updated.
- $\geq 70\%$  branch coverage is achieved, as required by SRS Section 6.4 (gcov/lcov).

## 5.1 Security Validation

Dedicated validation of the SRS Section 6.3 security requirements:

- SEC-01: Confirm passwords are stored only as a salted hash (e.g., SHA-256 with per-user salt); inspect users.dat to confirm no plaintext password is ever written.
- SEC-02: Confirm trains.dat, users.dat and bookings.dat are written only inside the application's own directory, with no access to system-level directories.
- SEC-03: Fuzz/boundary-test console and file-parsed inputs (oversized names, malformed records) to confirm length validation prevents buffer overflow or record injection.
- SEC-04: Verify a logged-in session is force-expired after 15 minutes of inactivity and requires re-authentication.
- SEC-05: Corrupt a data file deliberately and confirm the checksum/record-count check detects it and the system re-initialises the store safely (Safety Requirements, SRS 6.2) instead of crashing.

## 6. Test Environment

| Category             | Details                                                                                |
|----------------------|----------------------------------------------------------------------------------------|
| Operating Systems    | Windows 10/11 and Ubuntu 20.04 LTS (per SRS 2.4)                                       |
| Compiler / Toolchain | GCC 9+ / MSVC 2019+, C++17                                                             |
| Build System         | CMake 3.16+                                                                            |
| Unit Test Framework  | GoogleTest (gtest) 1.14+                                                               |
| Static Analysis      | SonarQube (or equivalent), integrated into Jenkins CI                                  |
| CI/CD                | Jenkins pipeline on the GitHub-hosted repository                                       |
| Coverage Tooling     | gcov / lcov, target $\geq 70\%$ branch coverage                                        |
| Test Data            | Seeded trains.dat, users.dat and bookings.dat with dummy trains, accounts and bookings |

## 7. Test Schedule

All dates are working targets against the Mini-Project Part-1 ETA of 28-Sep-2026.

| Milestone                                        | Target Date    | Owner                       |
|--------------------------------------------------|----------------|-----------------------------|
| Test plan & test case design complete            | 18-Sep-2026    | All members                 |
| Unit tests written alongside module code         | 18-22-Sep-2026 | Module owner (see Sec. 9)   |
| Integration & system test execution              | 23-25-Sep-2026 | All members                 |
| Security & performance validation (Sec 5.1, NFR) | 25-26-Sep-2026 | Pooja Koppad / Pavankumar H |
| Defect fixes & regression re-run                 | 26-27-Sep-2026 | All members                 |
| Final review, RTM sign-off, packaging            | 27-28-Sep-2026 | Pothineni Dinesh            |

## 8. Test Deliverables

- Software Test Plan (this document)
- Test Cases (Section 15) - manual and GoogleTest-automated
- Requirement Traceability Matrix (Section 13)
- Test execution logs and coverage reports (gcov/lcov)
- Defect log (tracked in the GitHub repository issues)
- Test Summary Report, submitted alongside the SRS and Architecture & Design Specification

## 9. Roles and Responsibilities

| Name                  | SRN           | Modules Owned                                        | Testing Responsibility                                                                           |
|-----------------------|---------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Pooja Koppad          | PES2UG24CS352 | AuthManager, AdminTrainManager, AdminFareManager     | Security validation (Sec 5.1); TC-01 to TC-03, TC-13; admin & auth test cases                    |
| Pooja Kanakappanavara | PES2UG24CS351 | TrainSearch, SeatInventory, BookingManager           | TC-04 to TC-07; search, seat-hold and booking/waitlist test cases                                |
| Pavankumar H          | PES2UG24CS345 | PaymentModule, CancellationManager, ReportGenerator  | TC-08 to TC-10, TC-13 (reports); payment, refund and performance (NFR) test cases                |
| Pothineni Dinesh      | PES2UG24CS354 | PNRService, HistoryService, TicketPrinter, FileStore | TC-11, TC-12, TC-14; PNR/history/e-ticket and data-integrity test cases; RTM & document assembly |

## 10. Risks and Mitigation

| Risk                                                            | Mitigation                                                                      |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| Module owner falls behind on unit tests before ETA              | Daily stand-up check-in; reassign a buddy tester from Section 9                 |
| GoogleTest / CMake environment differs across members' machines | Standardise via a shared CMakeLists.txt and documented setup steps in README.md |
| Security tests (Sec 5.1) require code not yet complete          | Prioritise AuthManager and FileStore completion first in the schedule           |
| Jenkins/SonarQube pipeline misconfiguration delays CI feedback  | Validate pipeline early with a trivial commit before feature work lands         |

## 11. Assumptions & Dependencies

- All four team members have a working C++17 toolchain (GCC/MSVC) and GoogleTest configured locally.
- The GitHub repository and Jenkins webhook are set up before integration testing begins.
- Test data (trains.dat, users.dat, bookings.dat seed files) is prepared before system-test execution starts.
- The SRS (Jackfruit Phase-1, 06-Sep-2026) remains the authoritative requirements source; any SRS change triggers an update to this test plan and the RTM.

## 12. Suspension & Resumption Criteria

**Suspend testing if:**

- The build fails to compile via CMake for more than 4 hours, or
- A defect blocks execution of more than 30% of planned test cases.

**Resume testing when:**

- The blocking defect is resolved and a green build is confirmed on Jenkins.

## 13. Test Case Management & Traceability

The Requirement Traceability Matrix (RTM) below maps each test case to the SRS requirement ID(s) it verifies and the owning module, extending the RTM in SRS Appendix C with test-level detail.

| Test Case ID | Requirement ID(s)                        | Module(s)                                            | Code File(s)                                   |
|--------------|------------------------------------------|------------------------------------------------------|------------------------------------------------|
| TC-01        | FR-01, FR-02                             | AuthManager                                          | Auth.cpp                                       |
| TC-02        | FR-03, FR-04, SEC-04                     | AuthManager                                          | Auth.cpp                                       |
| TC-03        | SEC-01                                   | AuthManager, FileStore                               | Auth.cpp, FileStore.cpp                        |
| TC-04        | FR-05, FR-06, FR-07                      | TrainSearch                                          | Search.cpp                                     |
| TC-05        | FR-08, FR-09                             | SeatInventory                                        | SeatInventory.cpp                              |
| TC-06        | FR-10, FR-11                             | BookingManager                                       | Booking.cpp                                    |
| TC-07        | FR-12, FR-13, FR-19                      | BookingManager, CancellationManager                  | Booking.cpp, Cancellation.cpp                  |
| TC-08        | FR-14, FR-15, FR-16                      | PaymentModule                                        | Payment.cpp                                    |
| TC-09        | FR-17, FR-18                             | CancellationManager                                  | Cancellation.cpp                               |
| TC-10        | NFR-01, NFR-02, NFR-03, NFR-04, NFR-05   | TrainSearch, BookingManager, FileStore               | Search.cpp, Booking.cpp, FileStore.cpp         |
| TC-11        | FR-20, FR-21, FR-22, FR-23, FR-24        | PNRService, HistoryService, TicketPrinter            | PNRService.cpp, History.cpp, TicketPrinter.cpp |
| TC-12        | SEC-05                                   | FileStore                                            | FileStore.cpp                                  |
| TC-13        | FR-25, FR-26, FR-27, FR-28, FR-29, FR-30 | AdminTrainManager, AdminFareManager, ReportGenerator | AdminTrain.cpp, AdminFare.cpp, Report.cpp      |
| TC-14        | SEC-02, SEC-03                           | FileStore                                            | FileStore.cpp                                  |

## 14. Test Metrics & Reporting

**Metrics collected:**

- % of planned test cases executed and % passed/failed
- Branch coverage (gcov/lcov), target  $\geq 70\%$  per SRS 6.4
- Defect count by severity and defect aging
- Requirement coverage - % of FR/NFR/SEC IDs with at least one passing test case

**Reports:**

- Per-pull-request Jenkins build/test/coverage/SonarQube summary
- Final Test Summary Report submitted with Part-1 deliverables

## 15. Detailed Test Cases

The following 14 test cases give functional and non-functional coverage across every feature area, satisfying the Part-1 requirement of at least 7-10 documented test cases (Deliverable 4).

| ID    | Title                                                                          | Req ID(s)                              | Preconditions                                                            | Steps                                                                                                                                                                                                                     | Expected Result                                                                                                                                                                                                                                           | Pri.   |
|-------|--------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC-01 | Successful registration & duplicate-email rejection                            | FR-01, FR-02                           | Application running, Main Menu shown.                                    | 1) Select Register. 2) Enter unique name/email/phone/password. 3) Repeat Register with same email.                                                                                                                        | First registration succeeds and account is created; second attempt is rejected with a duplicate-email error.                                                                                                                                              | High   |
| TC-02 | Login authentication and account lockout after 5 failed logins                 | FR-03, FR-04, SEC-04                   | A registered passenger account exists.                                   | 1) Select Login and enter correct email/password (verify access is granted only on correct credentials). 2) Re-attempt Login with correct email and wrong password 5 times consecutively.                                 | Step 1: passenger is authenticated and a session starts only on valid credentials (FR-03). Step 2: account is locked for 15 minutes after the 5th failed attempt and an audit event is logged (FR-04, SEC-04).                                            | Medium |
| TC-03 | Passwords never stored in plaintext                                            | SEC-01                                 | A passenger has registered.                                              | 1) Register a new account. 2) Inspect users.dat directly.                                                                                                                                                                 | Only a salted hash (e.g., SHA-256 with a per-user salt or equivalent approved hashing mechanism) is present; plaintext passwords are not stored.                                                                                                          | High   |
| TC-04 | Train search returns class/fare details, or a clear no-results message         | FR-05, FR-06, FR-07                    | Logged-in passenger at Main Menu.                                        | 1) Select Search Trains and enter a source/destination/date with scheduled trains; verify each matching train lists its available classes and per-class fares. 2) Repeat with a combination that has no scheduled trains. | Step 1: matching trains display with class-wise fares (FR-05, FR-06). Step 2: system displays a clear 'No trains found' message; no crash or blank output (FR-07).                                                                                        | Medium |
| TC-05 | Seat soft-lock expires after 5 minutes                                         | FR-08, FR-09                           | A train with available seats is found.                                   | 1) Select a train/class and confirm seat selection. 2) Wait beyond 5 minutes without completing passenger details.                                                                                                        | The soft lock expires, the seat(s) return to available inventory, and the session returns to Search Results.                                                                                                                                              | High   |
| TC-06 | Unique PNR generated on confirmed booking                                      | FR-10, FR-11                           | Seats held, passenger details entered.                                   | 1) Enter valid details for 1-6 travellers. 2) Confirm the booking.                                                                                                                                                        | A unique 10-digit PNR is generated and the booking is recorded with the correct traveller count.                                                                                                                                                          | High   |
| TC-07 | Waitlisted booking auto-upgrades on cancellation                               | FR-12, FR-13, FR-19                    | A train with zero seats available; one existing Confirmed booking on it. | 1) Create a new booking (expect Waitlisted). 2) Cancel the existing Confirmed booking on the same train/date/class.                                                                                                       | The cancelled seat is released back to seat inventory and triggers a waitlist check (FR-19); the waitlisted booking automatically upgrades to Confirmed status (FR-12, FR-13).                                                                            | Medium |
| TC-08 | Payment failure releases the seat hold; successful payment generates a receipt | FR-14, FR-15, FR-16                    | Passenger reaches the Payment screen with a fare calculated.             | 1) Submit a payment that is simulated to fail. 2) Retry with a payment simulated to succeed.                                                                                                                              | Step 1: booking is not confirmed, the seat hold is released back to inventory, and the user returns to Seat Selection (FR-14, FR-15). Step 2: booking is confirmed and a payment receipt with a unique transaction ID is generated and persisted (FR-16). | High   |
| TC-09 | Refund calculated per cancellation-fee slab                                    | FR-17, FR-18                           | A Confirmed booking exists with departure >24h away.                     | 1) Cancel the booking via PNR before the 24-hour cutoff. 2) Repeat with a booking cancelled within 24h of departure.                                                                                                      | Refund reflects the 25% fee tier when cancelled >24h before departure and the 50% fee tier when cancelled within 24h, matching SRS Appendix B.2 parameters.                                                                                               | High   |
| TC-10 | Core performance targets are met (search, booking, startup, file I/O, memory)  | NFR-01, NFR-02, NFR-03, NFR-04, NFR-05 | Dataset seeded with 200 trains; application built in a representative    | 1) Measure Search Trains response time. 2) Measure time from seat hold to booking confirmation. 3) Measure application startup                                                                                            | Search results $\leq 2s$ (NFR-01); booking transaction $\leq 3s$ (NFR-02); startup $\leq 3s$ (NFR-03); file I/O $\leq 200ms$ (NFR-04); memory footprint $\leq 128MB$ (NFR-05).                                                                            | Medium |

| ID    | Title                                                                  | Req ID(s)                                | Preconditions                                                                                            | Steps                                                                                                                                                                                                                                                                                          | Expected Result                                                                                                                                                                                                                                                                          | Pri.   |
|-------|------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|       |                                                                        |                                          | Release configuration.                                                                                   | time (launch to Main Menu). 4) Measure booking/user/train file read-write time. 5) Measure peak memory usage during an active session.                                                                                                                                                         |                                                                                                                                                                                                                                                                                          |        |
| TC-11 | PNR status, booking history and e-ticket reflect current booking state | FR-20, FR-21, FR-22, FR-23, FR-24        | A Confirmed booking exists with a known PNR, alongside at least one past booking for the same passenger. | 1) Query PNR status. 2) Open Booking History and filter by upcoming/past, then sort by journey date. 3) Generate the e-ticket for the PNR and save/print it as a local file.                                                                                                                   | PNR status shows Confirmed with correct seat/coach details (FR-20); history correctly filters upcoming/past and sorts by date (FR-21, FR-22); e-ticket contains matching PNR, train, seat, fare and passenger details (FR-23) and is saved/printed successfully as a local file (FR-24). | Medium |
| TC-12 | Corrupted data file is detected and safely re-initialised              | SEC-05, Safety (6.2)                     | A valid bookings.dat exists.                                                                             | 1) Manually corrupt bookings.dat (e.g., truncate a record). 2) Launch the application.                                                                                                                                                                                                         | System detects the checksum/record-count mismatch on load and re-initialises the store without throwing an unhandled exception or losing unaffected data.                                                                                                                                | High   |
| TC-13 | Administrator can manage schedules, fares, quotas and generate reports | FR-25, FR-26, FR-27, FR-28, FR-29, FR-30 | Admin credentials configured separately from passenger accounts; at least one train schedule exists.     | 1) Log in via Admin Login using separate admin credentials. 2) Add/edit a train schedule record. 3) Update the fare for a travel class and the seat quota for a category. 4) Generate an occupancy report for a specified train/date. 5) Generate a revenue report for a specified date range. | Admin login succeeds only with valid admin credentials (FR-25); schedule changes persist (FR-26); fare and quota updates persist (FR-27, FR-28); the occupancy and revenue reports display correctly computed values (FR-29, FR-30).                                                     | High   |
| TC-14 | Data files stay within the app directory and inputs are validated      | SEC-02, SEC-03                           | Application installed in its standard directory; console and file-parsed inputs available for testing.   | 1) Trigger read/write operations on trains.dat, users.dat and bookings.dat and confirm the file paths used. 2) Submit an oversized/malformed console input (e.g., an excessively long name) and a manually crafted malformed record in a data file.                                            | All data files are written only within the application's own directory, with no access to system-level directories (SEC-02); oversized or malformed inputs are rejected or length-checked with no buffer overflow or malformed-record injection (SEC-03).                                | High   |

## 16. Approvals

| Role                          | Name | Signature / Date |
|-------------------------------|------|------------------|
| Team Lead                     |      |                  |
| Course Instructor / Evaluator |      |                  |